

CASEPATH: AN AGENTIC, PROCESS-FIRST ARCHITECTURE FOR DETERMINING EVIDENCE REQUIREMENTS

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ABSTRACT

Insurance claims are a useful testbed for AI decision systems because the evidence needed to resolve a case depends on where that case lies in a rule-governed process. A single fact can activate a different branch, create new obligations, and change which documents are justified. Yet many LLM-based approaches bypass this structure and predict document checklists directly from case text. Such requests may look reasonable while being weakly grounded: the system can ask for the right-looking document for the wrong reason. We introduce CASEPATH, an agentic, process-first architecture in which the source-defined process—not the language model—controls evidence acquisition. Agents read authoritative sources, place the case on the applicable process path, identify active or unresolved requirements, and request only the evidence needed to satisfy them. As evidence arrives, the case state is updated and the required checklist changes accordingly. We test this dependency using matched cases that differ in one branch-defining fact. Across 27 held-out pairs, CASEPATH makes 15 false-positive document changes, versus at least 27 for every comparison system. This turns document prediction into evidence planning that is selective, traceable, and auditable.

1 INTRODUCTION

A customer reports a burglary. A stolen watch is separately scheduled under an insurance policy. That fact activates a requirement to establish its value, through a purchase receipt or an expert valuation. Remove the fact and the requirement disappears. Leave it uncertain and the next step is a clarification question. A receipt may therefore be necessary, irrelevant, premature, or one of several sufficient alternatives within the same claim category. The requirement depends on the active obligation, not simply on the document’s association with theft.

A direct predictor can learn this dependency. Its checklist, however, does not reveal whether a request follows from the applicable rule or from a plausible document prior. CASEPATH makes the dependency executable:

source → process node → obligation → required fact → evidence capability → document route. (1)

The model reads passages and assesses case conditions; a controller derives the requests from active obligations. An unresolved condition yields a question. A false condition contributes no branch-specific demand. Each request retains its source, condition and evidentiary purpose, and new evidence updates the state from which the plan is derived (Figure 1). This leads to a testable prediction: *resolving process obligations before predicting documents should make evidence plans more selective when a case changes.*

We test selectivity directly rather than equating it with document recall. Study A changes one branch-defining fact in each of 36 matched case pairs and scores the justified change in their checklists. On the 27 held-out pairs, CASEPATH makes 15 false-positive changes, against 27 for Direct, 41 for Graph as context and 52 for Evidence-first. Against Direct, it avoids twelve false-positive changes while recovering one fewer correct change; its signed-change F_1 is 0.667 versus 0.588. The improvement is concentrated in procedural branches whose required documents are poorly suggested by the claim category alone. The family analysis identifies both these strengths and the semantic errors that still generate wrong requests.

Source rule: a scheduled valuable requires evidence of its value

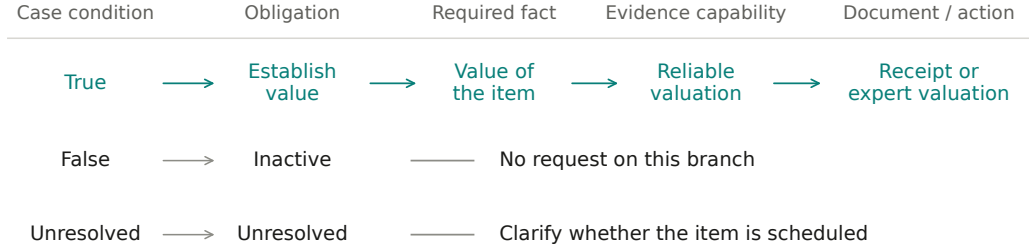


Figure 1: **The active branch controls the request.** Schematic of the motivating example. A source condition determines whether an obligation is active, inactive or unresolved. An active obligation connects a required fact to an evidence capability and its document routes; uncertainty calls for clarification. The version-specific rules for choosing routes and judging sufficiency are defined in Section 2.

Study B tests the same principle on the complete 150-claim tenancy release. It exposes a native interface failure, then separates literal-request agreement from a retrospective evaluation at the model’s recorded assessed state. That second comparison isolates inherited process scope. On the 90 protected-family cases, scope raises reference-chain precision from 0.318 to 0.732; the conservative paired benefit remains +0.103 with unavailable pairs penalized. On the same completed development cases, both controllers recover 259 valid requests, while scope reduces total requests from 592 to 275. The mechanism is selective acquisition: keep the evidence that active obligations require and suppress demands inherited from inapplicable processes.

The contribution is the coupling of a source-grounded document-planning object to an executable controller, tested at the level of branch-induced changes and full request lists. Compiled procedures and evidence ledgers already separate semantic judgement from control (Kim, 2026; Yu et al., 2026); we examine the consequence for evidence acquisition. The release joins the method, benchmarks, recorded outputs, evaluator and workbench so that a reader can inspect why a request was made and reproduce what the experiments measured.

2 METHOD: SOURCE-GROUNDED PROCESS STATE CONTROLS EVIDENCE

2.1 THE PLANNING OBJECT

A source specification K records process nodes, conditions, obligations, required facts, evidence capabilities and document routes over a closed catalogue $\mathcal{D}$. A capability states what must be shown; a route is a subset of $\mathcal{D}$ that can establish it, and alternative routes remain alternatives. Case state $s = (z, E)$ pairs the assessment z of source conditions with the recorded evidence E . A versioned planner returns the requested documents, a justified next action and the justification chains,

$$(D_t, a_t, J_t) = \Pi_K(s_t), \quad s_{t+1} = U(s_t, e_{t+1}), \quad (2)$$

where U admits a new observation or correction. Two properties hold by construction. *Non-prematurity*: an unresolved condition can emit a question but never a branch-specific document. *Provenance closure*: D_t is a projection of active chains, so no document can enter a plan without an obligation, a fact, a capability and a route behind it.

2.2 AGENTIC ROLES, AND WHERE CODE TAKES OVER

Semantic work is done by model calls in fixed roles: offline, extracting quote-anchored propositions from admitted passages, synthesising the macro process, refining propositions into guarded evidence rules and mapping capabilities onto the catalogue; online, deciding for each guard whether a case satisfies it. Every role returns a structured judgement, never a checklist. Code owns activation, projection, route selection, next-action priority and the justification record (Appendix A): a model

may say what a passage means or whether a case meets a condition, but it cannot add a document to the plan.

2.3 STUDY A PLANNER: GUARDED RULES OVER A CLOSED CATALOGUE

The frozen Study A specification was compiled from 197 exact passages of Swiss insurance law and public household-insurance policies and forms (Fedlex, CSS, AXA, Generali, Simpego and the Swiss Insurance Association), with recorded URLs and content hashes. Extraction yields 308 operational propositions. A deterministic check requires every proposition to be either assigned to an evidence rule or explicitly classified as non-evidentiary; the frozen representation has 98 evidence rules, 13 non-evidentiary classifications, 87 case guards and a catalogue of 33 document types. Completeness holds over this extracted inventory only.

Each rule r has a local applicability guard g_r , required capabilities $\mathcal{C}_r$ and, for each capability c , routes $\mathcal{R}_c$. The interpreter assigns $z_g(x) \in \{\text{T}, \text{F}, \text{U}\}$ to guard g for case x . A true or false verdict is accepted only with an exact quotation from that same case; otherwise it reverts to unresolved. Absence of evidence is unresolved, never false. Each guard is decided in its own call, with case packets batched only as an execution optimisation, so every branch decision is auditable on its own. Let $A(x)$ be the unconditional rules and the rules whose guards are true, and let $\mathcal{R}_g^{\text{conf}}$ be the confirmation routes mapped for a guard that is true in the narrative but not yet established by admitted evidence. With held document types H , the plan is

$$D(x, H) = \left[\bigcup_{r \in A(x)} \bigcup_{c \in \mathcal{C}_r} \bigcup_{\rho \in \mathcal{R}_c} \rho \cup \bigcup_{g \text{ true, unconfirmed}} \bigcup_{\rho \in \mathcal{R}_g^{\text{conf}}} \rho \right] \setminus H. \quad (3)$$

An unresolved guard yields a clarification question; a false guard contributes nothing; no generative step runs after the verdicts. Each request retains its rule, fact, must-show statement and source spans. Two boundaries of this version matter for reading the results: Equation 3 unions mapped routes, so V5 neither selects one sufficient alternative nor judges whether a held document is adequate for a specific capability; and the gate is rule-local, so changing macro-process edges while keeping rules and verdicts fixed cannot change D . “Active path” in Study A means the case-conditioned applicability structure, not traversal of every edge.

2.4 STUDY B CONTROLLER: APPLICABILITY, ACQUISITION AND ADEQUACY

The complete-corpus controller compiles three public tenancy templates deterministically into one representation shared by every arm; it performs no model-based source induction. Its assessment separates predicate truth, document presence, individual adequacy and joint adequacy. On the applicability graph,

$$q_s(z) = \phi_s(z) \wedge_3 J_s(\{q_p(z)\}_{p \in \text{pa}(s)}), \quad a_o(z) = q_{\text{scope}(o)}(z) \wedge_3 \phi_o(z), \quad b_o(z) = a_o(z) \wedge_3 \psi_o(z), \quad (4)$$

where J_s is the declared all/any join over parents (roots use the empty all-join, which is true), $\wedge_3$ is false if any input is false, true if all are true and unresolved otherwise, a_o is applicability and ψ_o a separate acquisition permission. Only $b_o = \text{T}$ permits an immediate request. For capability c , a route ρ is satisfied only when its required evidence is present and adequate and the joint assessment is sufficient; write this state $\rho_c(E)$, with missing or inadequate documents $M_{c\rho}$ and unknown presence or adequacy forming review items $V_{c\rho}$. The controller selects

$$\rho_c^* = \arg \min_{\rho \in \mathcal{R}_c}^{\text{lex}} (\mathbf{1}\{\rho_c(E) \neq \text{T}\}, |M_{c\rho}|, |V_{c\rho}|, \text{sort}(M_{c\rho}), \text{id}(\rho)), \quad (5)$$

per capability rather than by global minimisation, and requests $\bigcup_c M_{c\rho_c^*}$ over eligible obligations. Inactive obligations add no demand; unresolved eligibility produces conditional candidates and questions; an empty route set is an evidence gap, not licence to invent a document. The next action follows a fixed priority: resolve questions, review evidence, request a document, review an active gap, then choose an eligible substantive action. This controller extends the Study A planning object with sufficiency semantics; it is not a repair of V5, and no comparison between the runtimes is claimed (Appendix B).

3 STUDY A: INTERVENING ON THE ACTIVE BRANCH

Construction. We select 9 branch concepts whose truth changes at least one evidence requirement under the sources: a scheduled valuable at or above the policy threshold, a bicycle among the stolen items, SIM-card theft with misuse, an identifiable third-party causer, a formal expert procedure, a repair above the consent threshold, a written demand with a forfeiture period (*Nachfrist*), recovery of stolen goods, and insurer declination. Each concept is instantiated in four neutral household-theft contexts, giving 36 pairs and 72 case units. The members of a pair differ in one statement: for the watch above, “No stolen item is separately scheduled in the policy with an insured value of CHF 1,000 or more” becomes “A stolen watch is separately scheduled in the policy with an insured value of CHF 4,000”. A reference contract authored from the same sources before any system was run fixes the acceptable signed changes; where the wording admits alternatives (receipt *or* expert valuation) they remain alternatives, and both pre-hidden amendments are preserved with the earlier versions. One context per concept forms the development split (9 pairs, 18 units); the remaining three contexts (27 pairs, 54 units) stayed sealed until the method, benchmark, analysis code and gate were hash-frozen, and were read once. The held-out split therefore tests context robustness for known rule concepts, not unseen-rule or unseen-policy transfer.

Falsifiers built into the benchmark. The benchmark tests whether gains can be explained by static predictions, request volume or branch-independent changes (Kaushik & Lipton, 2018; Sugawara et al., 2018; McCoy et al., 2019; Bowman & Dahl, 2021; Yuan et al., 2024). (i) *Input-free control*: a constant signed-set predictor fitted on eight branch families and evaluated on the ninth has maximum held-out micro- F_1 0.0; the nine target signatures are distinct (3.17 bits of entropy over 12 signed document atoms). (ii) *Wrong-pairing permutation*: CASEPATH’s outputs are held fixed and reassigned to other branch families under 8 cyclic shifts and 5,000 seeded derangements; a system whose changes are not branch-specific keeps its score. (iii) *One-shot read with a preregistered gate*: eight conditions (Table 4) were fixed before the held-out execution, including a deliberately strong broad-superiority requirement.

Scoring. For a pair (x^-, x^+) , a system’s signed change is

$$\Delta(D) = \{+d : d \in D(x^+) \setminus D(x^-)\} \cup \{-d : d \in D(x^-) \setminus D(x^+)\}. \quad (6)$$

The route-aware scorer compares Δ with every acceptable reference realisation and keeps the best under a frozen lexicographic rule (Appendix A); requesting the union of two alternatives is not credited as correct. True positives, predictions and references are pooled over the 27 pairs into micro precision, recall and F_1 . Spurious changes are unjustified additions or withdrawals; missed changes are required ones not made; an exact pair reproduces the entire signed set. These count changes between two checklists, not total requests in a claim. Uncertainty uses 10,000 whole-family bootstrap draws; pairwise contrasts use the exact $2^9 = 512$ family sign-flip test with Holm adjustment.

Comparators and budget. *Direct* predicts the document set from the case and source material. *Graph as context* additionally receives the induced process and evidence-rule representation but still generates the set in one step. *Evidence-first* derives evidence needs, retrieves twelve passages by keyword matching, and then predicts documents. All arms use the same model setting (openai/gpt-5.6-terra), provider restriction, temperature zero, source snapshot, catalogue and case texts; the held-out receipts record 54, 54, 108 and 87 calls. Inputs and model conditions are matched; compute is not (Appendix A).

4 STUDY A: SELECTIVE BRANCH ADAPTATION

Fewer unjustified changes. Table 1 reports the held-out comparison. CASEPATH predicts 39 signed changes for 33 reference changes: 24 correct, 15 spurious, 9 missed. *Direct* predicts 52 (25 correct, 27 spurious), *Graph as context* 72 (31, 41) and *Evidence-first* 68 (16, 52). CASEPATH thus makes 44.4%, 63.4% and 71.2% fewer spurious changes than the three alternatives while keeping recall 0.727, and it reproduces the whole signed set on 12 of 27 pairs against 9, 4 and 2. *Graph as context* recovers the most reference changes (31 of 33) at the cost of 41 spurious ones. In this comparison, providing the graph as context improves recall but does not impose the controller’s

Method	Signed changes			Agreement	
	Correct	Spurious	Missed	Exact pairs	F_1 [95% CI]
Direct	25	27	8	9/27	0.588 [0.404, 0.759]
Graph as context	31	41	2	4/27	0.590 [0.471, 0.716]
Evidence-first	16	52	17	2/27	0.317 [0.239, 0.400]
CASEPATH	24	15	9	12/27	0.667 [0.435, 0.880]

Table 1: **Held-out signed-change evaluation** (27 pairs, 33 reference changes). Correct, spurious and missed count signed changes. Exact counts pairs whose entire signed set is reproduced; F_1 includes its 95% family-cluster bootstrap interval. These are errors in changes between two checklists, not requests in a complete claim.

restriction on what may be requested. The contrast with Direct is precise: twelve fewer false-positive changes for one fewer correct change.

A retrospective partition sharpens the interpretation: 9 of CASEPATH’s 15 spurious changes concern documents outside the benchmark’s branch-governed set, compared with all 27, 41 and 52 comparator errors. Process control reduces this form of checklist drift; it does not eliminate it. The partition does not identify universally required baseline documents, which the signed-change references do not specify.

How much does the result generalize? Family-cluster intervals are wide: F_1 0.667 [0.435, 0.880] for CASEPATH against [0.404, 0.759] and [0.471, 0.716] for the two strongest comparators, with paired differences $[-0.253, +0.420]$ and $[-0.199, +0.333]$ that include zero; only the contrast with Evidence-first excludes it ($[+0.110, +0.565]$). The broad-superiority gate fails on three of its eight conditions (Table 4): margins +0.078 and +0.076 fall short of 0.10, precision 0.615 falls short of 0.70, and the Holm-adjusted family-swap p -values are 0.797, 0.797 and 0.129. The count reduction supports selectivity on these pairs; the preregistered criterion for broad superiority is not met.

Branch specificity. Holding CASEPATH’s outputs fixed and pairing them with the wrong branch families collapses signed-change F_1 from 0.667 to a maximum of 0.083 over the 5,008 evaluated reassignments (mean 0.011; add-one randomisation $p = 1.996 \times 10^{-4}$). The changes are specific to the intervened branch. This does not identify the internal representation as the only possible cause; a family-specific lookup would also pass. Two structural facts are separate from accuracy: all 39 of 39 predicted changes are attributed in the retained records to a changed guard, including the incorrect ones, and 27 of 33 reference changes (81.8%) are reachable through a source chain in the frozen representation. Attribution is an invariant of the architecture, not evidence that a mapping was right; reachability says a route existed, not that it was emitted.

Procedural branches explain the advantage. Figure 2 shows the structure the aggregate hides. CASEPATH is exact on all three held-out contexts of four concepts (bicycle, declination, recovery, *Nachfrist*), reaches 0.667 on scheduled valuables and SIM-card theft, 0.500 on repair consent, and scores zero on third-party causation and the formal expert procedure. The zeros are systematic rather than random: an alternative route is over-requested where one would suffice, a condition is attached at the wrong semantic granularity, or an adjacent procedural guard activates the wrong confirmation document. The comparators are strong where the plausible claimant document is the right answer (third party, repair consent) and weak where the justified change is a procedural artefact that a checklist prior does not suggest (declination, *Nachfrist*); per-concept values are in Appendix A. Finally, a change metric is blind to a document omitted or over-requested in both members of a pair, and these cases are rule units rather than whole claims; Study B therefore scores complete plans on whole cases.

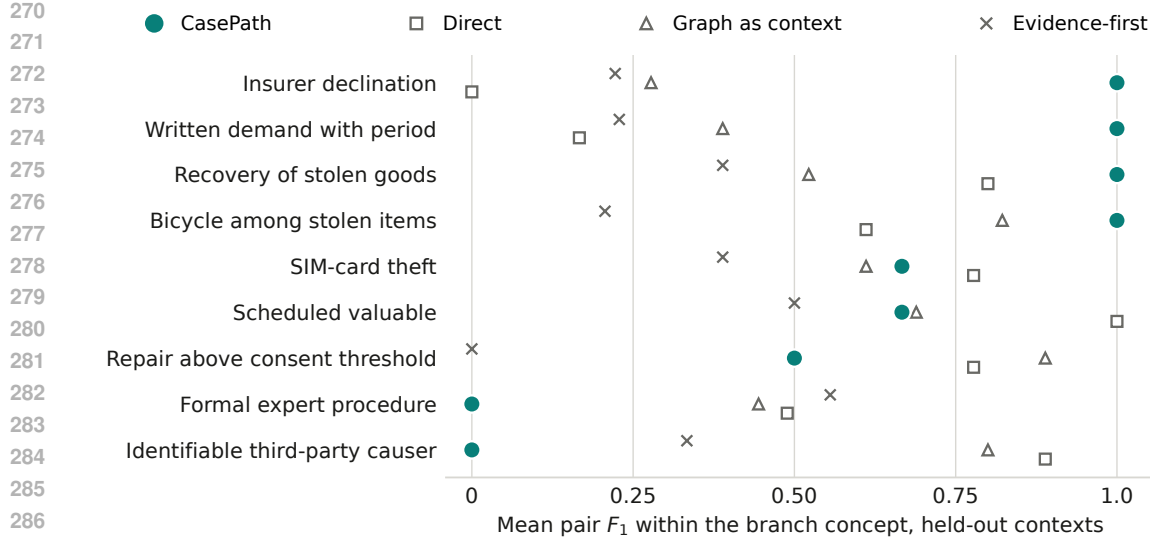


Figure 2: **Where the semantics holds and where it fails.** Mean pair F_1 per branch concept on the held-out contexts: the marker is CASEPATH, the square, triangle and cross identify Direct, Graph as context and Evidence-first. Small vertical offsets separate coincident values. CASEPATH is exact on four concepts where the comparators are not, and scores zero on two where they succeed. Concepts are ordered by CASEPATH’s score, so the failures sit at the bottom beside the comparators that solved them.

5 STUDY B: COMPLETE CLAIMS AND THE VALUE OF PROCESS SCOPE

A matched complete-corpus comparison. Study B evaluates all 150 released synthetic tenancy claims across 28 families in three domains: defects, mould and heating; termination; and rent increase. Development contains 60 claims in 11 families; the protected split contains 90 claims in 17 disjoint families. Five learned pipelines share the compiled public representation, observable packet, source access, model alias and budget: 2 calls per cell, medium reasoning, 4,096 completion tokens per call, no tools or resampling. CASEPATH emits an assessment that code compiles into the artefact; Direct, Document-first, Process-context and Rule-first generate the artefact. Compiled-equivalent and local-scope controls reuse CASEPATH’s assessments. This compares complete pipelines under matched information and budget.

Registered primary and separately specified analyses. All 558 generated artefacts fail the original native interface: the compiler’s condition names are not bound in the reference scenario. The frozen policy penalizes all 1,050 cells, including 492 failed or blocked outputs; 0 of 12 protected-split practical targets are met. The primary report and failures remain unchanged. Two retrospective analyses answer narrower questions: whether literal requests agree with reference sets, and whether inherited scope improves a plan evaluated at its recorded assessed state. Neither replaces the registered result.

Realized request quality across all pipelines. The first analysis scores literal immediate requests with the original reference and counting functions, without interpreting guards. All 558 generated lists are scoreable; failed outputs retain zero quality scores and an unnecessary-fraction penalty of one. Table 2 shows the complete comparison. CASEPATH produces 66 lists on protected families, with failure-penalized F_1 0.677 against 0.458 for the strongest comparator, Process-context. The difference includes output availability: the conventional paired advantage is +0.219, but the conservative contrast, assigning -1 whenever either endpoint is unobserved, is -0.673 . This supports better realized output under the budget, not reliable paired superiority over the learned alternatives.

Making the assessed state explicit. The second analysis specializes each saved artefact to its own authenticated assessment. A condition known true activates its entities; a false condition does not;

Pipeline	Generated lists		Checklist $F_1 \uparrow$		Unnecessary $\downarrow$	
	Dev.	Prot.	Dev.	Prot.	Dev.	Prot.
CASEPATH	44	66	0.657	0.677	0.296	0.268
Direct	19	30	0.295	0.310	0.687	0.688
Document-first	17	24	0.247	0.249	0.732	0.737
Process-context	27	43	0.407	0.458	0.570	0.538
Rule-first	28	42	0.433	0.450	0.545	0.540
<i>Dependent controls: same recorded CasePath assessments</i>						
Compiled equivalent	44	66	0.657	0.677	0.296	0.268
Local-scope ablation	44	64	0.420	0.411	0.669	0.682

Table 2: **Complete-corpus request comparison.** Retrospective agreement of literal request lists with reference sets. Scores weight domains equally, families equally within domains and cases equally within families. Failed outputs remain in the denominator; “Unnecessary” includes their unit penalty. Dependent controls reuse the same recorded assessments.

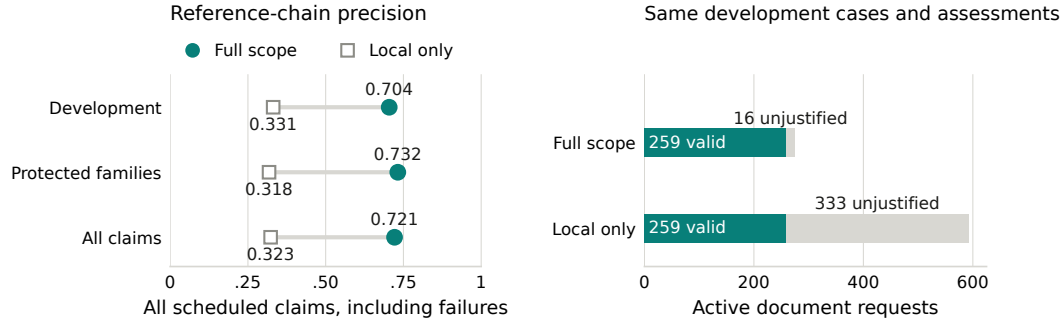


Figure 3: **Inherited scope removes unjustified demand.** Left: reference-chain precision in the retrospective current-case evaluation, including every scheduled case and failure penalty. Right: on the same 44 completed development cases, both controllers retain 259 valid requests; enclosing scope reduces total requests from 592 to 275. The assessment and source representation are fixed.

unresolved entities remain recorded as unresolved and are not asserted active. The resulting current-case snapshot can be scored without substituting the reference scenario for the model’s beliefs. The original requests and all failed cells are retained. Only CASEPATH and its two dependent controls have the required saved assessments. Every available snapshot scores successfully, and none excludes a literal immediate request. This interface evaluates current-case structure, not the program’s response to hypothetical branch changes (Appendix E).

The benefit is selectivity at fixed assessment. Figure 3 isolates inherited domain scope: an obligation must satisfy the enclosing process conditions as well as its local guard. On protected families, native valid-chain precision rises from 0.318 to 0.732; the conservative paired benefit remains positive at +0.103 even after penalizing unavailable pairs. The corresponding unnecessary fraction falls from 0.682 to 0.268. These are complementary views of the same requests, not independent effects. On the same 44 completed development cases, both controllers recover 259 valid requests, while total requests fall from 592 to 275. Scope therefore removes unjustified demands without sacrificing those recovered obligations. The compiled-equivalent control matches CASEPATH exactly. The intervention establishes the value of inherited domain applicability; it does not isolate arbitrary graph topology. Native chain matching checks agreement with reference decision–fact–capability–document links, not exact-source entailment: provenance-chain inheritance is zero in both controllers.

The family label records the original target-custody split: observable inputs had been inspected during product work, historical target non-access is not established because the producer’s complete I/O audit is missing, and targets are now released. These are finite-corpus descriptive analyses, not fresh hidden confirmation. Costs are \$89.06 for the study and \$89.66 including earlier activity; the

retrospective analyses add no provider calls. Full endpoints, raw counts and paired contrasts are in Appendix D and Appendix E.

6 RELATED WORK

Executable rules and adaptive case state. LegalRuleML represents norms with their source associations (OASIS LegalRuleML Technical Committee, 2021); Catala makes legislative computation executable (Merigoux et al., 2021); DMN formalises decisions and CMMN guard-activated case work (Object Management Group, 2024; 2016); Guard-Stage-Milestone models and DCR reference models of law connect data-driven activation to compliance (Vaculín et al., 2011; López et al., 2020); proactive enforcement derives actions from active obligations (Basin et al., 2024); and insurance software conditions document lists on service definitions (ServiceNow, 2026). This established separation of rules and state underlies CASEPATH; we couple it to language-model assessments and a measurement of the resulting evidence-planning errors.

Procedure extraction and process-guided agents. PAGED extracts procedural graphs from documents (Du et al., 2024); BREX benchmarks business-rule flows with executable grounding (Yang et al., 2026); Guideline2Graph compiles clinical guidelines into provenance-preserving decision graphs (Kilic et al., 2026); Graus (2026) evaluate structured representations for legal decision models; CANONIC frames governance as compilation (Hadley, 2026). FlowBench and WorFBench evaluate workflow-guided planning and workflow generation (Xiao et al., 2024; Qiao et al., 2025); Procedural Graphs localise an agent to an active node (Lu et al., 2026); ReAct interleaves reasoning and action (Yao et al., 2023). Our evaluation instead asks whether changing a case fact changes exactly the evidence justified by that branch.

Obligation-controlled acquisition. The closest precedents are Kim (2026), whose evidence ledger activates branch-specific knowledge obligations while a model judges semantic warrant and code manages closure, and Yu et al. (2026), who compile SOP constraints into verification routines with evidence-bearing returns and an active execution frame. CASEPATH studies their application to a document-planning object: applicability separated from acquisition permission, presence from capability-specific and joint adequacy, alternative routes kept as alternatives, and a benchmark that tests the consequence downstream. Neither system is reproduced as an arm, and no superiority over them is claimed.

Grounding, evidence and information seeking. GraphCompliance aligns policy and context graphs for regulatory assessment (Chung et al., 2025); CLER scores litigation evidence lists and the facts they prove (Bi et al., 2026); InfoGatherer selects retrieval and questions by evidential uncertainty (Taranukhin et al., 2026); verified quotation and chain-of-verification make support inspectable (Menick et al., 2022; Dhuliawala et al., 2024). Tool-use benchmarks score whether the right function was called (Li et al., 2023; Patil et al., 2025), whereas the present studies evaluate the case-specific justification for an evidence request.

Benchmark validity. Partial-input baselines, annotation artefacts and shortcut learning motivate the input-free control and the wrong-pairing permutation (Kaushik & Lipton, 2018; Poliak et al., 2018; Gururangan et al., 2018; McCoy et al., 2019; Sugawara et al., 2018; Geirhos et al., 2020; Bowman & Dahl, 2021; Yuan et al., 2024). Study A’s references are authored from sources rather than adjudicated by a model, avoiding the preference leakage that arises when judged and judging models coincide (Li et al., 2026); work on evaluation validity (Thomas et al., 2026) and critiques of legal benchmarks that reward outcome prediction over practitioners’ work (Medvedeva & McBride, 2023; Lou & Shin, 2026) motivate signed document changes and emitted requests as the measured objects.

7 RELEASE: ONE PLANNER FOR RESEARCH AND PRODUCT

A research method and a claims interface can silently diverge. The Study A release packages the frozen representation, propositions, snapshot, method manifest, benchmark identity and model configuration into a hash-bound method pack; the client cannot inject rules, mappings or targets, and

hash drift makes the endpoint fail closed. The product calls the same guard interpreter and planner as the experiment and exposes guard states, chains, questions, gaps, requests, exact source spans and the next action. Replaying all 72 benchmark units with the recorded guard decisions reproduces the research document sets, guard states and next actions on every unit and all 36 signed pair changes, with every rendered quotation an exact substring of its passage: implementation identity conditional on recorded assessments, not fresh-inference agreement or deployment evidence. The workbench organises six roles (Facts, Orchestration, Source Integrity, Process, Evidence, and Audit and Readiness) around one case record, and a newly admitted observation triggers recomputation through Equation 2. The release also contains benchmark versions, predictions, receipts, analysis code, the parity verifier, and Study B’s corpus, producer, evaluator, contract and every recorded cell. Both the registered primary failure and the separate request-only diagnostic reproduce offline from the exact released reference containers. Reference targets and authored teaching states are displayed separately from model predictions; neither is additional benchmark evidence.

8 DISCUSSION AND LIMITATIONS

The practical value of process control is selectivity with an inspectable reason. In Study A, CASEPATH removes false-positive changes while retaining nearly as many correct changes as Direct. The benefit is largest on procedural branches, where a familiar claimant document is a poor substitute for the particular notice or confirmation that the rule requires. Study B’s current-case scope intervention establishes a related design choice: an obligation must inherit the conditions of its enclosing process, not merely satisfy its own local condition. A system builder can inspect and test both dependencies before asking a customer for another document.

The evidence has specific limits. Study A tests known concepts in new contexts and additions rather than justified withdrawals, uses unequal computation across pipelines, and does not meet its pre-registered broad-superiority gate. Its source-derived references contain author judgements about which documents establish a fact, rather than expert consensus; a deployed system would also need to select the governing policy instead of combining rules from multiple insurers. Study B tests complete synthetic claims under matched budgets, but its native interface failure leaves counterfactual graph correctness unvalidated. Its retrospective current-case analysis isolates scope within dependent controls; the broader request comparison measures realized output under failures. Neither study measures customer effort, expert agreement, autonomous acquisition or performance in another jurisdiction.

9 CONCLUSION

Evidence planning benefits from making the reason for a request executable. CASEPATH represents the dependency from source condition to obligation, fact, capability and document, and lets case state determine which demands are active. The controlled comparison finds fewer unjustified checklist changes; the current-case intervention shows that inherited scope removes unjustified demand while retaining the same recovered obligations in development. These results support a concrete design principle: derive evidence demands from explicit, case-conditioned obligations, and validate both the requests and the process semantics that produce them. The released benchmarks, controller, traces and offline reproduction make that principle inspectable and testable.

REPRODUCIBILITY STATEMENT

The release contains, for Study A, the frozen snapshot, propositions, rules, document map, interpreter, planner, benchmark versions, all predictions, provider receipts, analysis code, the preregistered gate and the product replay verifier, and for Study B its corpus, compiled representation, producer, evaluator, analysis contract and every recorded cell, including failures. Every number in this paper is generated from those files by a script that records the file hash and JSON pointer behind each value. The release commands recompute Study A and all three Study B analyses from the recorded outputs and compare them with the preserved measurements without network or provider access (Appendices A and D). Reproducing scores from recorded outputs is distinct from regenerating model outputs, which requires provider access and is not deterministic.

ETHICS STATEMENT

CASEPATH is a research architecture for evidence planning, not legal advice or an automated coverage decision. The studies use public-source snapshots and synthetic cases with no personal data. Real deployment requires policy and jurisdiction scoping, professional review, privacy controls and a mechanism for correction and appeal.

AI USE STATEMENT

Generative AI tools were used in hypothesis formulation, methodological critique, literature discovery, implementation and testing, synthetic-case construction, diagnostic interpretation, and drafting and editing of text and figures. The reproducibility materials distinguish actual executions, engineering fixtures, archived reports and author judgements; failed results and version differences are retained. The authors are responsible for the final scientific claims, citations, measurements and artefacts.

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A STUDY A: BENCHMARK AND ANALYSIS DETAILS

Branch concepts. The 9 concepts are: a scheduled valuable at or above the source-defined threshold; a bicycle or e-bike among the stolen items; SIM-card theft with a misuse claim; an identifiable third-party causer; selection of formal expert proceedings; contemplated repairs above the consent threshold; a written demand with a forfeiture warning and period; recovery of, or news about, stolen items; and insurer declination. Each has four neutral contexts; a pair changes only the branch statement.

Signed-change scoring. For pair i the scorer evaluates each acceptable signed reference realisation $G \in \mathcal{G}_i$ against the prediction $P_i = \Delta(D_i)$ and selects the maximum of the lexicographic tuple $(F_1(P_i, G), |P_i \cap G|, \text{Prec}(P_i, G), \text{Rec}(P_i, G), -|G|, \text{sort}(G))$, whose last component is the frozen deterministic tie-break. Precision and recall use zero for empty denominators; F_1 is one when prediction and reference are both empty. Micro scores pool the selected realisation counts across the 27 pairs; family means average pair F_1 within each of the nine concepts. The family bootstrap resamples nine whole families with replacement on each of 10,000 draws (seed 20260917), retaining all three held-out contexts of each selected family. The exact family-swap test enumerates 512 sign assignments of the nine family mean differences; the three one-sided p -values (0.436, 0.398, 0.043) receive Holm adjustment (0.797, 0.797, 0.129). The wrong-pairing diagnostic uses 8 cyclic shifts and 5,000 sampled derangements with add-one score $(1 + N_{\geq})/(1 + 5,008)$; the reported maximum is over this evaluated collection, not all derangements.

Per-concept results. Table 3 lists the held-out mean pair F_1 for every branch concept and arm.

Branch concept	Direct	Graph ctx.	Evid.-first	CASEPATH
Bicycle among stolen items	0.611	0.822	0.206	1.000
Insurer declination	0.000	0.278	0.222	1.000
Formal expert procedure	0.489	0.444	0.556	0.000
Recovery of stolen goods	0.800	0.522	0.389	1.000
Written demand with period (Nachfrist)	0.167	0.389	0.229	1.000
Repair above consent threshold	0.778	0.889	0.000	0.500
Scheduled valuable	1.000	0.689	0.500	0.667
SIM-card theft	0.778	0.611	0.389	0.667
Identifiable third-party causer	0.889	0.800	0.333	0.000

Table 3: Held-out mean pair F_1 by branch concept.

Preregistered gate. Table 4 lists the eight conditions fixed before the one-shot read and their outcomes.

Preregistered held-out gate	Result
F_1 margin ≥ 0.10 against every comparator	fail
Holm-adjusted family-swap $p \leq 0.05$ against every comparator	fail
Signed-change precision ≥ 0.70	fail
Predicted/gold change ratio in $[0.75, 1.50]$	pass
Predicted changes attributed to a changed guard ≥ 0.90	pass
Reference changes reachable through a source chain ≥ 0.80	pass
Correct pairing above the wrong-pairing 97.5th percentile	pass
No requested document without a source chain	pass

Table 4: Preregistered held-out gate, fixed before the one-shot read. The provenance, volume and pairing conditions pass; the broad-superiority conditions do not.

Frozen record and development separation. The method freeze casepath.theft-method-freeze/5.0.0 binds source bytes, benchmark v3 and both amendments, the proposition-complete refinement, document mapping, interpreter V4, planner V3, model, provider and temperature settings, analysis code, the development split, the matched comparators’ raw predictions and

the focused method tests. Development used 9 pairs (18 units), one context per concept, and its pre-declared admission checks authorised exactly one held-out read of the remaining 27 pairs (54 units) under a separately hash-bound execution freeze. After the held-out result, the method was marked locked and no scientific file was adapted from held-out outcomes.

Calls and recorded costs. Table 5 lists the held-out provider receipts per arm.

Arm	Calls	Prompt tokens	Completion tokens	Max completion	Cost (USD)
Direct	54	2,900,625	45,678	14,000	4.83
Graph as context	54	5,345,421	42,663	14,000	8.41
Evidence-first	108	259,890	66,450	14,000/18,000	1.06
CASEPATH	87	1,005,211	243,891	10,000	5.44

Table 5: Held-out provider receipts, one logged attempt per call, model `openai/gpt-5.6-terra`, temperature 0. Evidence-first allows 18,000 completion tokens for extraction and 14,000 for planning; CASEPATH allows 10,000 per guard batch; Direct and Graph as context allow 14,000 per call. Costs are provider-reported charges, exclusive of source preparation; the arms are not compute-matched.

Product-parity contract. The parity verifier loads the installed hash-bound method pack and the real product service, replays all 72 units with the recorded guard decisions, compares guard verdicts, document sets and next actions, compares all 36 pair deltas through the product diff surface, and checks that every rendered quotation is an exact substring of its source passage. A successful HTTP response or a plausible demo is not treated as evidence of implementation identity.

Descriptive error origin. The retrospective error partition uses the union of reference documents in unit annotations and all acceptable signed changes, with each arm’s original scorer-selected realization. It preserves the benchmark’s signed-change counts. The bicycle-voucher amendment was reflected in acceptable deltas but not two auxiliary vocabulary fields; including both definitions leaves the error counts unchanged. A single sentence intervention need not change only one inferred guard, so the observed absence of withdrawals is not claimed as a monotonicity result. The release includes the partition script, all counts and this field-consistency note.

B VERSIONED CORRESPONDENCE BETWEEN THE TWO STUDIES

The shared interface is source specification, case assessment, evidence planning, state replacement and a justified next action. It does not assert identical knowledge, algorithms, estimands or validation across the two implementations. Study A is bound to `casepath.theft-method-freeze/5.0.0`, interpreter V4 and evidence overlay V3; Study B is bound to the compact native producer and its `obligation_control_v1` and `evidence_demand_v1` controller. The native public template union is not a transfer of the theft source pack, and the theft pairs are not transformations of the native claim roster. V5 judges local guards and projects every mapped route on active rules plus mapped confirmation evidence for true unconfirmed guards, minus held document types; it implements neither capability-specific nor joint adequacy. The native controller distinguishes applicability from acquisition, unknown from missing evidence, and per-capability route selection from route union. These are substantive version differences, and no benchmark comparison between the runtimes is claimed. Study A’s preparation extracted propositions, synthesised and refined rules and mapped capabilities with model calls and retained source-only corrections; its reported held-out costs exclude that preparation. Study B’s compilation is deterministic and shared by all arms.

C STUDY B: NATIVE MEASUREMENTS AND ANALYSIS CONTRACT

Native endpoints. Native critical-evidence recall is criticality-weighted coverage of active evidence obligations through any complete permitted document alternative. Evidence-obligation completeness additionally requires valid decision–fact–capability–document relations. Valid-chain precision counts requested items with such a chain over all requested items; its empty-set value is

Binding	Study A: theft-method-freeze/5.0.0	Study B: native-150 compact controller
Knowledge	Source-extracted propositions, macro graph, refined local evidence rules and mapped routes	Common deterministic compilation of all three public tenancy templates
Online gate	Rule-local true/false/unresolved guard; macro edges do not gate requests	Inherited applicability scopes, obligation condition and separate acquisition permission
Evidence state	Held document types; true-guard confirmation; union of mapped routes	Presence, capability/route adequacy, multi-document joint adequacy; per-capability route selection
State update	Recompute requests/next action from a changed packet; paired edit, not acquired evidence outcome	Recompute from a new assessment; benchmark measures one planning realization
Population	9 development + 27 held-out pairs; nine shared concepts, four contexts; 72 units total	60 development/11 families + 90 protected/17 families; 150 claims, three tenancy domains
Learned inference	Guard-major batched V5; 1-call Direct, 1-call representation baseline, 2-call Evidence-first	Five 2-call singleton pipelines; two additional controls reuse CasePath state
Endpoint	Route-aware signed additions/withdrawals, 27-pair micro scores	Complete native artifact and separate actual immediate-checklist metrics
Analysis	Historical family bootstrap, exact family swaps and wrong-family diagnostic; broad gate failed	Equal domain/family/case finite contrasts; family-deletion sensitivity, no significance claim
Parity	Recorded-answer replay over 72 units and 36 deltas	Separate implementation; no inherited V5 or six-role/live-app parity claim

Table 6: A common source-to-state-to-evidence interface does not make the two implementations, populations or measurements interchangeable. Study B extends the evidence question rather than retrospectively re-evaluating V5.

one and must be read beside coverage. Wrong-branch and premature requests, document states, node and edge scores, branch and path probes and source locators retain their native contracts; locator agreement is not semantic entailment, and probe accuracy is not a direct measure of inferred case-predicate accuracy. The emitted endpoint counts unique immediate request identities without activating candidate conditions from the reference scenario, and is available only when the reference has a unique allowed document set for every active obligation, complete consistent document states and the public label mapping. For predicted and reference sets P, G : $F_1 = 2|P \cap G|/(|P| + |G|)$ and unnecessary fraction $U = |P \setminus G|/|P|$; both-empty F_1 , empty-prediction precision and empty-reference recall are one, and empty-prediction burden is zero. Failed cells use adverse frozen unit penalties instead: zero for higher-is-better rates, one for lower-is-better rates. Unavailable reference labels never become empty sets or silently shrink a denominator.

Estimation. For a split or the complete corpus, each scheduled case has weight $\omega_i = 1/(3 m_{d(i)} n_{f(i)})$, where m_d is its domain’s family count and n_f its family’s scheduled case count. Conventional differences use adverse-imputed unit scores; conservative benefit-oriented differences contribute -1 whenever either producer endpoint is unobserved. Every single-family deletion is reported with remaining weights renormalised within the affected domain; these ranges are composition sensitivity, not confidence intervals. A comparator is counted as beaten only when all three conservative contrasts strictly exceed their boundaries on the complete protected population under the original references and integrity checks; otherwise the report labels the outcome as not met, a partial practical signal, or not evaluable.

Execution and failure classes. Each cell records its arm, case, state and either the validated artefact or the retained provider output with the reason validation failed. Unknown relation endpoint: a candidate artefact relation references a node identity that does not exist in the artefact. Truncated or non-terminal: the completion reached the 4,096-token limit or ended without a terminal outcome. Invalid codebook index: a reference into the shared public codebook that does not resolve. Contra-

dictory presence state: a document marked both present and absent, or present with an inconsistent state. Blocked dependency: a dependent control whose CASEPATH source cell failed. Every failure retains its charge; a repaired version can never replace a failed cell in the frozen comparison.

D POST-HOC REQUEST-ONLY DIAGNOSTIC

The registered wrapper scores emitted requests only after a native receipt has no evaluation failures. A common expression-namespace mismatch therefore suppresses request scoring for every generated artefact. The post-hoc diagnostic changes that availability rule alone: it reads the unchanged `request_mode=now` items, maps their labels with the original public mapping, and applies the original emitted-set scorer. It never evaluates a candidate condition against a reference state. All original artefacts, producer failures, native failures, primary penalties and costs remain separately available.

All 150 references yield an eligible unique document set. The 558 generated lists receive observed request-only scores; 492 failed or blocked outputs receive the original adverse checklist penalties, with no invented requests. Domain-family-case weighting is unchanged. For benefit direction $b_m \in \{-1, 1\}$, the conventional paired contribution is $b_m(s_{i,CP,m} - s_{i,a,m})$; the conservative contribution is the same quantity when both request-only endpoints are observed and -1 otherwise. A graph's native failure does not make an independently scoreable request list unobserved in this diagnostic. No practical threshold, significance test or confidence interval is applied.

Development: 60 claims						
Pipeline	Lists	Correct/req.	Missed	Cond.	Prec.	Recall
CASEPATH	44	259/275	41	811	0.704	0.635
Direct	19	114/116	16	23	0.313	0.281
Document-first	17	95/103	22	84	0.268	0.233
Process-context	27	163/169	23	15	0.430	0.394
Rule-first	28	171/174	21	11	0.455	0.414
Compiled equivalent	44	259/275	41	811	0.704	0.635
Local-scope ablation	44	259/592	41	560	0.331	0.635
Protected families: 90 claims						
Pipeline	Lists	Correct/req.	Missed	Cond.	Prec.	Recall
CASEPATH	66	384/385	54	1195	0.732	0.645
Direct	30	187/200	14	42	0.312	0.310
Document-first	24	142/144	17	55	0.263	0.238
Process-context	43	270/280	13	16	0.462	0.456
Rule-first	42	261/265	15	25	0.460	0.443
Compiled equivalent	66	384/385	54	1195	0.732	0.645
Local-scope ablation	64	376/853	52	827	0.318	0.627
Combined: 150 claims						
Pipeline	Lists	Correct/req.	Missed	Cond.	Prec.	Recall
CASEPATH	110	643/660	95	2006	0.721	0.641
Direct	49	301/316	30	65	0.312	0.298
Document-first	41	237/247	39	139	0.265	0.236
Process-context	70	433/449	36	31	0.451	0.434
Rule-first	70	432/439	36	36	0.458	0.432
Compiled equivalent	110	643/660	95	2006	0.721	0.641
Local-scope ablation	108	635/1445	93	1387	0.323	0.630

Table 7: **Request counts and coverage.** Lists counts the generated artefacts that contribute raw counts. Correct/req., Missed and Cond. sum only those artefacts; Cond. counts documents deferred conditionally, not immediate requests. Precision and recall are full-population, failure-penalized weighted scores, and therefore differ from the ratios of the displayed raw counts. Dependent controls reuse the same assessments.

Comparator	Endpoint	Conventional	Conservative
Direct	F_1	+0.367	-0.758
Direct	Unnecessary fraction	+0.419	-0.744
Document-first	F_1	+0.429	-0.788
Document-first	Unnecessary fraction	+0.469	-0.785
Process-context	F_1	+0.219	-0.673
Process-context	Unnecessary fraction	+0.270	-0.657
Rule-first	F_1	+0.227	-0.671
Rule-first	Unnecessary fraction	+0.272	-0.650

Table 8: Protected-family paired request-only contrasts. Positive values favor CASEPATH; the unnecessary-fraction difference reverses its sign so that less burden is positive. The conservative rule assigns -1 when either endpoint is unobserved. All endpoint contrasts, all families and every family-deletion result are included in the machine-readable report.

The boundary is semantic, not just syntactic. The public compiler introduces 50 condition symbols shared by all arms. None of the emitted symbols lies outside that namespace. The native expression language evaluates them against the contract’s scenario, which does not provide the needed assignment. An authored, target-blind counterexample makes the difference explicit: the product controller returns unresolved for an absent condition; the native parser raises a missing-variable error. Assigning null instead makes the native Boolean evaluation false, while the controller still returns unresolved. Without a public state correspondence, a rename, default or injected predicted state would change meaning. No such correction is made.

A missing document becomes a request only when its obligation is active

	Arrears procedure	Family-home protection
Case condition	Arrears: true	Family home: unresolved
	↓	↓
Obligation	Cure notice with warning	Separate spouse notice
	↓	↓
Evidence state	Notice missing	Notice missing
	↓	↓
Document plan	Request the cure notice	Keep spouse notice conditional

The recorded next action resolves applicability before acquisition.

Figure 4: **A recorded controller decision.** In the first completed CASEPATH development case in sorted cohort order, arrears is true and family-home status unresolved. The cure notice is requested now; the spouse notice remains conditional. This shows two routes within the recorded plan, not an accepted native graph: its exported predicate namespace encounters the same interface failure as the other generated artefacts. The release includes the complete case trace and selection rule.

Execution evidence and historical access. The primary scoring task authenticates both complete prediction inventories before reading references. The original producer terminated without its complete I/O audit; historical non-access to protected targets is therefore not established retrospectively. Scoring, the compatibility diagnosis and the request-only analysis are separately recorded, with original failures preserved. All original target containers are subsequently released as exact bytes under their recorded data license. This version is now a public benchmark and cannot serve as newly hidden confirmation.

Offline reproduction. The anonymous artifact contains all 1,050 prediction and failure cells, the exact 150 reference containers, original native evaluator, reporting code, source commitments and request-only wrapper. Network-disabled replay reproduces both primary phase reports, the complete

primary report, every request-only row and its descriptive report by exact canonical JSON comparison, without a numerical tolerance. This reproduces scoring from recorded model outputs; it does not regenerate those outputs or provide the remote model checkpoint.

E RETROSPECTIVE ASSESSED-STATE EVALUATION

Evaluation object. The assessed-state interface binds every compiler predicate to the complete three-valued assessment saved with that prediction. Each original condition is evaluated under this predicted state. Known-active entities enter a current-case snapshot; unresolved entities remain explicitly recorded and are not asserted active. Counterfactual branch predicates are omitted from this snapshot, so branch-probe accuracy and full-program acceptance are not endpoints. The envelope retains the original artefact, assessment and every activation decision. Reference states are never used to decide predicted activation. Native reference graphs, scoring functions and entity matchers are unchanged.

Negation swaps true and false; conjunction is true only if all children are true and false if any is false; disjunction uses the dual rules. Each control retains its own graph and requests, bound to its saved CASEPATH assessment. All 110 compiled-equivalent snapshots and scores match CASEPATH. No immediate request is filtered out. The release preserves the derivatives, identity checks and execution receipts.

Endpoints and interpretation. Native valid-chain precision counts active requested document entities linked through a reference-aligned decision–fact–capability–document chain, divided by active requested documents. Critical-evidence recall weights covered reference facts by their supplied criticality. Obligation completeness counts reference obligations with a valid matching chain. Unnecessary fraction counts requests that are not necessary under the reference. In this corpus the valid-chain and necessary-document sets coincide, so precision and unnecessary fraction are complementary, and critical-evidence recall equals obligation completeness. They are not independent confirmations. Exact-source entailment is not established: provenance-chain inheritance scores zero, and all-object source support is unavailable for this predicate-excluding interface.

Endpoint	Development		Protected		Combined	
	Full	Local	Full	Local	Full	Local
Node precision	0.523	0.263	0.536	0.256	0.531	0.258
Node recall	0.542	0.542	0.544	0.526	0.543	0.532
Node F_1	0.530	0.353	0.539	0.344	0.536	0.347
Edge precision	0.676	0.378	0.679	0.368	0.678	0.372
Edge recall	0.374	0.374	0.381	0.370	0.379	0.372
Edge F_1	0.480	0.376	0.487	0.369	0.485	0.371
Critical-evidence recall	0.635	0.635	0.645	0.627	0.641	0.630
Obligation completeness	0.635	0.635	0.645	0.627	0.641	0.630
Valid-chain precision	0.704	0.331	0.732	0.318	0.721	0.323
Unnecessary fraction	0.296	0.669	0.268	0.682	0.279	0.677
Source-chain inheritance	0.000	0.000	0.000	0.000	0.000	0.000
Scored cells	44	44	66	64	110	108
Valid / all requests	259/275	259/592	384/385	376/853	643/660	635/1445

Table 9: **Current-case structure and evidence at fixed assessment.** Full denotes CASEPATH with inherited domain scope; Local uses only the obligation’s local scope. All scheduled cases contribute with the original failure penalties and domain–family–case weights. Raw counts describe available outputs; protected and combined output coverage differs across controllers. Compiled-equivalent reproduces Full exactly.

The conservative paired benefit uses the benefit-oriented per-case difference when both endpoints exist and -1 otherwise. Chain-precision benefits are $+0.112$ in development, $+0.103$ in protected families and $+0.107$ overall; critical-recall benefits are -0.261 , -0.289 and -0.278 . This supports selectivity at comparable coverage. The report includes every family and paired contrast. These are descriptive finite-corpus quantities, without post-hoc significance tests or confidence intervals.